1. Clinical Question & Core Hypothesis

Clinical Question

Intravesical BCG provokes local immune organization (TILs/TLS) in NMIBC, yet many patients with apparent local immune activation still fail to achieve durable BCG responses. Why does local immune organization sometimes not translate into effective tumor control and durable clinical response? 2) Tier 1 paths used: I built on Core Path 1 (BCG → trial-derived biomarkers), Core Path 2 (BCG → PD-L1 → bTMB) and Core Path 3 (ADNP mRNA detectable in CTCs as a liquid-biopsy signal). These together provide a mechanistic bridge from local immune organization (histology) → immune-suppressive checkpoint induction (PD-L1) → systemic molecular readout of tumor programs/dissemination (ADNP in CTCs). 3) Tier 2 evidence supporting the claim: MSI/neoantigen context and Immunoscore concepts support that antigenicity and immune-contexture modulate immune efficacy; AI-based signatures offer pathways to integrate histology and circulating signals; methylation markers (SDC2/TFPI2) illustrate cross-modality, non-invasive biomarker feasibility. These support a multi-modal view where local histologic immune structure can be functionally overridden by tumor-intrinsic or systemic programs. 4) Why this endpoint: I chose a therapy_response endpoint (BCG response proxy) because the mechanistic puzzle concerns why BCG fails in some immune-infiltrated tumors and because trials and liquid biopsies are naturally linked to response monitoring—making therapy response the most directly relevant clinical outcome. 5) Deliberate exclusions: I excluded OS/PFI to keep the hypothesis therapy-focused and temporally proximal to BCG; I excluded broad panel/model-driven statements (e.g., combining many genes into a predictive signature) to keep the claim mechanistic and testable. Exploratory paths (CSF/neuro-immune, systemic metabolic markers, PSA) were omitted from the core hypothesis due to weak direct mechanistic ties to NMIBC/BCG response?

Working Hypothesis

In NMIBC treated with intravesical BCG, TLS-like lymphoid aggregates reflect a local anti-tumor immune attempt but do not guarantee durable response when tumors upregulate PD-L1, which functionally suppresses TLS-primed T cell activity. I hypothesize that: (i) high intratumoral PD-L1 within TLS-rich NMIBC correlates with poor durable response to BCG, indicating a functionally immunosuppressive TLS state; (ii) this immune-evasive phenotype is echoed systemically by elevated ADNP mRNA in circulating tumor cells (CTCs), representing tumor-intrinsic programs of immune evasion and potential dissemination; (iii) a multi-modal model that integrates TLS architecture, PD-L1 expression, ADNP CTC signal, and systemic tumor metrics (bTMB, MSI, SDC2/TFPI2 methylation, Immunoscore-like scores, and AI-derived integration signatures) will outperform TLS alone in predicting BCG response; (iv) patients categorized as TLS-positive with high PD-L1 and ADNP+ CTCs will have the worst responses, whereas TLS-positive with low PD-L1 and ADNP will display improved outcomes, and (v) combining BCG with PD-1/PD-L1 blockade in the high-risk subgroup will enhance durable responses, as supported by ex vivo organoid-T cell assays and preclinical NMIBC models.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PD-L1, ADNP
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PD-L1: PD-L1 → bTMB) and Core Path 3 (ADNP mRNA detectable in CTCs as a liquid-biopsy signal).

ADNP: ADNP mRNA detectable in CTCs as a liquid-biopsy signal).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph
Secondary	AI-based signatures	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: ; AI-driven integrative signatures; ex vivo organoid–T cell co-cultures to test PD-1/PD-L1 blockade effects; and murine NMIBC models to validate therapeutic synergy. Limitations include potential sampling bias, detection sensitivity for CTCs, and the need to distinguish causation from association in ADNP signals.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.